



Project Report :Student Mental Health Detection

Course Name: Machine Learning

Course Code: CSE

Submitted to:

Raisa Fairouz

Lecturer,

Department of Computer Science & Engineering

Metropolitan University, Sylhet.

Submitted By:

Fatema Jannat Ema

ID: 232-115-019

Jannatul Ferdous

ID:232-115-040

Section: 59(A)

Department of Computer Science & Engineering

Metropolitan University, Sylhet

Date of submission: 14-08-2026

ABSTRACT

Student mental health is an important issue that can be influenced by academic workload, sleep patterns, social media usage, financial stress, family support, physical activity, and previous mental health experiences. This project applies machine learning techniques to analyze student-related information and predict students' mental health risk categories.

The dataset used in this project contains 2,000 student records and 17 columns. It includes demographic, academic, lifestyle, social, and mental-health-related attributes. The analysis begins with exploratory data analysis, including dataset inspection, descriptive statistics, missing-value analysis, target distribution analysis, and visualization. The project includes visualizations for mental health risk distribution, sleep hours, financial stress, and correlations between numerical features.

The data preprocessing stage removes the **Student_ID** identifier, handles missing values, scales numerical features, and applies one-hot encoding to categorical features. The dataset is divided into 80% training data and 20% testing data using stratified sampling. Five classification algorithms are evaluated: Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and Support Vector Machine (SVM).

The experimental results show that Logistic Regression achieved the highest accuracy of 75.50%, followed by SVM with 71.75%, Random Forest with 69.25%, KNN with 57.00%, and Decision Tree with 56.75%. Therefore, Logistic Regression is the best-performing model among the algorithms tested in this project.

INTRODUCTION

University students may experience different levels of mental health risk due to academic pressure, sleep patterns, study habits, social media usage, screen time, financial stress, family support, physical activity, diet, and previous mental health history.

The objective of this project is to analyze these factors and use machine learning to classify students into different mental health risk categories.

The target variable in the dataset is:

Mental_Health_Risk

It contains three categories:

- Low
- Medium
- High

Therefore, this project is formulated as a multi-class classification problem.

Objectives

The main objectives of this project are:

1. To explore the student mental health dataset.
2. To understand the structure and characteristics of the data.
3. To identify missing values and statistical properties.
4. To visualize important patterns in the dataset.
5. To preprocess numerical and categorical features.
6. To train multiple machine learning classification algorithms.
7. To evaluate and compare model performance.
8. To identify the best-performing classification model.

Importance of the Project:

Mental health is an important aspect of student well-being and academic life. Data analysis can help identify patterns among academic, lifestyle, social, and financial factors.

A machine learning model can provide a data-driven approach for classifying mental health risk categories. However, the prediction produced by this project should not be considered a medical or psychological diagnosis.

Scope

The project focuses on the information available in the provided student mental health dataset. It includes:

- Data collection
- Data exploration
- Data cleaning
- Exploratory Data Analysis
- Data preprocessing
- Machine learning classification
- Model evaluation
- Model comparison

DATA OVERVIEW

The dataset used in this project is:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Student_ID	Age	Gender	Year_of_St	Sleep_Hou	Study_Hou	GPA	Social_Me	Screen_Tir	Physical_A	Financial_L	Family_Su	Attendanc	Part_Time	Diet_Quali	Previous_H	Mental_Health_Risk	
2	STU0001	22	Male	3	6.6	17.6	2.91	6.9	7.5	Moderate	8.5	4	78	No	Poor	No	High	
3	STU0002	21	Male	4	5.6	22.4	2.92	7.6	11.5	Low	4.5	5.3	85.4	Yes	Average	No	Medium	
4	STU0003	22	Male	5	7.1	28.1	2.68	6.5	8.9	Low	4.8	6.1	79.5	No	Average	No	Medium	
5	STU0004	24	Male	1	4.5	14.8	2.23	5.5	7.6	High	3.7	8	73.9	No	Average	Yes	High	
6	STU0005	20	Female	1	6.7	17.2	2.69	4.3	7	Moderate	8	7.4	99.8	Yes		No	Medium	
7	STU0006	20	Female	4	6.8	18.6		8.5	11.8	High	7	3.9		Yes	Average	No	Medium	
8	STU0007	24	Male	2	7.7	15	2.77	8.8	10.2	Low	6.1	7.7	100	No	Good	No	Low	
9	STU0008	23	Non-binary	3	7.9	20.5	2.65	6.1	7.6	Low		6.5	78.9	No	Average	Yes	Medium	
10	STU0009	20	Female	3	6.2	28.6	2.42	3.9	7.2	Moderate	5.3	8.4	72.6	No	Good	No	Low	
11	STU0010	22	Male	3	6.6	20.5	2.28	6.9	7.4	Moderate	7.3	5.7	92.7	No	Average	No	Medium	
12	STU0011	20	Male	5	6	31.3	2.94	4.8	6.6	Low	8.8	3.6	82.6	No	Good	No	High	
13	STU0012	20	Female	5	5.6	27.2	2.81	5.2	11.3	Moderate	5.4	8.7	87.8	No	Poor	Yes	Medium	
14	STU0013	22	Female	4	7.1	22.1	2.33	6.7	8	Moderate	5.2	6.9	74.5	Yes	Poor	Yes	High	
15	STU0014	17	Female	2	5.4	17	2.4	6.6	10.5	Low	7		84.1	Yes	Average	Yes	High	
16	STU0015	17	Male	4	6.3	23.3	2.83	4.5	6.3	Low	8.7	5.7	77.4	No	Good	No	High	
17	STU0016	20	Male	1	5.7	17.4	2.08	5.9	9	Moderate	7.2	6.8	82	No	Average	No	Medium	
18	STU0017	19	Male	1	7.4	20.5	2.96	6.6	9.9	Moderate	6.4	5.3	84.4	No	Average	No	Medium	
19	STU0018	22	Male	4	8.8	25.4	2.76	6	9.9	Moderate	5.4	9.4	88.2	No	Poor	No	Low	
20	STU0019	19	Male	2	5.4	22.7	2.94	6	8.3	Moderate	8.2	5.1	92	No	Average	No	High	
21	STU0020	18	Male	1	4.8	11.1	2.01	7.8	12.7	Low	7.7	5.6	83.7	Yes	Average	No	High	
22	STU0021	24	Female	1	6.7	13.3	2.06	6.3	8.1	High	6.3	4.7	85.5	No	Good	No	Medium	
23	STU0022	21	Female	4	6.8	21.1	2.47	6.1	8.9	High	7.2	5.5	84.1	No	Average	Yes	Medium	
24	STU0023	21	Male	3	7	22.3	2.63	2.4	5.2	Moderate	6.9	7.2	82.7	No	Average	Yes	Medium	
25	STU0024	18	Female	3	6.3	19.6	2.41	5.6	6.8	Low	3.3	7	87.8	No	Poor	No	Low	
26	STU0025	20	Male	1	6.6	17.9	2.43	7	9.2	Moderate		9.2	83.8	No	Poor	Yes	Medium	
27	STU0026	21	Female	4		20.3	2.71	5.6	8.3	Moderate	7.2	5.1	85.7	Yes	Average	No	High	
28	STU0027	19	Male	4	7.0	24.0	2.72	5.1	7.1	High	8.7	6.1	80.0	No	Average	Yes	Medium	

Figure :Dataset

student_mental_health.csv

The dataset contains 2,000 observations and 17 columns.

The variables are:

Feature	Type
Student_ID	Categorical/Identifier
Age	Numerical
Gender	Categorical
Year_of_Study	Numerical
Sleep_Hours	Numerical
Study_Hours_Per_Week	Numerical
GPA	Numerical
Social_Media_Hours	Numerical
Screen_Time_Hours	Numerical
Physical_Activity	Categorical
Financial_Stress	Numerical
Family_Support	Numerical
Attendance_Rate	Numerical
Part_Time_Job	Categorical
Diet_Quality	Categorical
Previous_Mental_Health_History	Categorical
Mental_Health_Risk	Target

The dataset contains 8 float variables, 2 integer variables, and 7 object/categorical variables. We collect data from kaggle.

<https://drive.google.com/file/d/1K0GSK0tM66C9RBt01imGHbV4mLXgePmT/view?usp=sharing>

DATA CLEANING AND PREPROCESSING

Missing Value Analysis

The dataset contains missing values in several columns. The missing-value analysis produced the following results:

Feature	Missing Values
Sleep_Hours	62
GPA	38
Social_Media_Hours	58
Financial_Stress	91
Family_Support	54
Attendance_Rate	47
Diet_Quality	69

The remaining columns do not contain missing values.

Removing Identifier

Student_ID is an identifier rather than a meaningful predictive feature. Therefore, it is removed before model development.

The target variable is:

Mental_Health_Risk

and all remaining relevant variables are used as input features.

Numerical Feature Processing

Numerical features are processed using two steps:

1. Median Imputation
2. Standard Scaling

The median strategy is used to replace missing numerical values, while **StandardScaler** standardizes numerical features.

Categorical Feature Processing

Categorical features are processed using:

1. Most-Frequent Imputation
2. One-Hot Encoding

The encoder uses **handle_unknown="ignore"** so that unseen categories do not create errors during transformation.

EXPLORATORY DATA ANALYSIS

Descriptive Statistics

The dataset was analyzed using descriptive statistics.

Important values include:

Feature	Mean	Minimum	Maximum
Age	21.13	17	29
Year_of_Study	2.63	1	5
Sleep_Hours	6.76	3.3	10
Study_Hours_Per_Week	20.0	2.4	38.3

GPA	2.56	1.5	3.66
Social_Media_Hours	6.01	2.1	10
Screen_Time_Hours	8.55	3.3	14
Financial_Stress	5.96	1.4	10
Family_Support	6.52	1.7	9.9
Attendance_Rate	85.0	54.4	100
	8		

These statistics are generated directly from the dataset using `data.describe()`.

Mental Health Risk Distribution

The target variable contains three classes:

Mental Health Risk	Students
High	680
Medium	660
Low	660
Total	2,000

The classes are relatively balanced, with 680 High-risk students and 660 students in each of the Medium and Low categories.

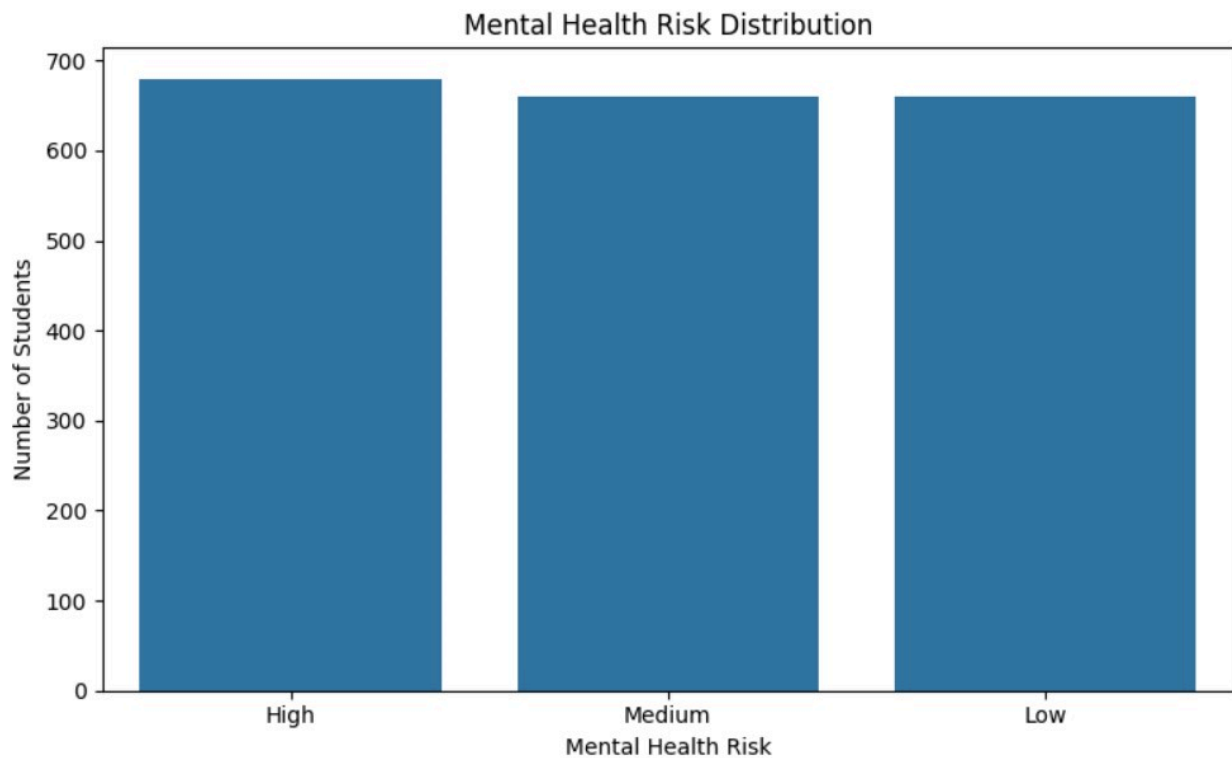


Figure 1. Distribution of students across Low, Medium, and High mental health risk categories.

The graph shows that the three target classes have a relatively similar number of observations. This is useful because the dataset does not have a severely imbalanced target variable.

Sleep Hours Analysis

A boxplot is used to analyze sleep hours across the three mental health risk categories. The notebook specifically includes a Sleep Hours Analysis

visualization.

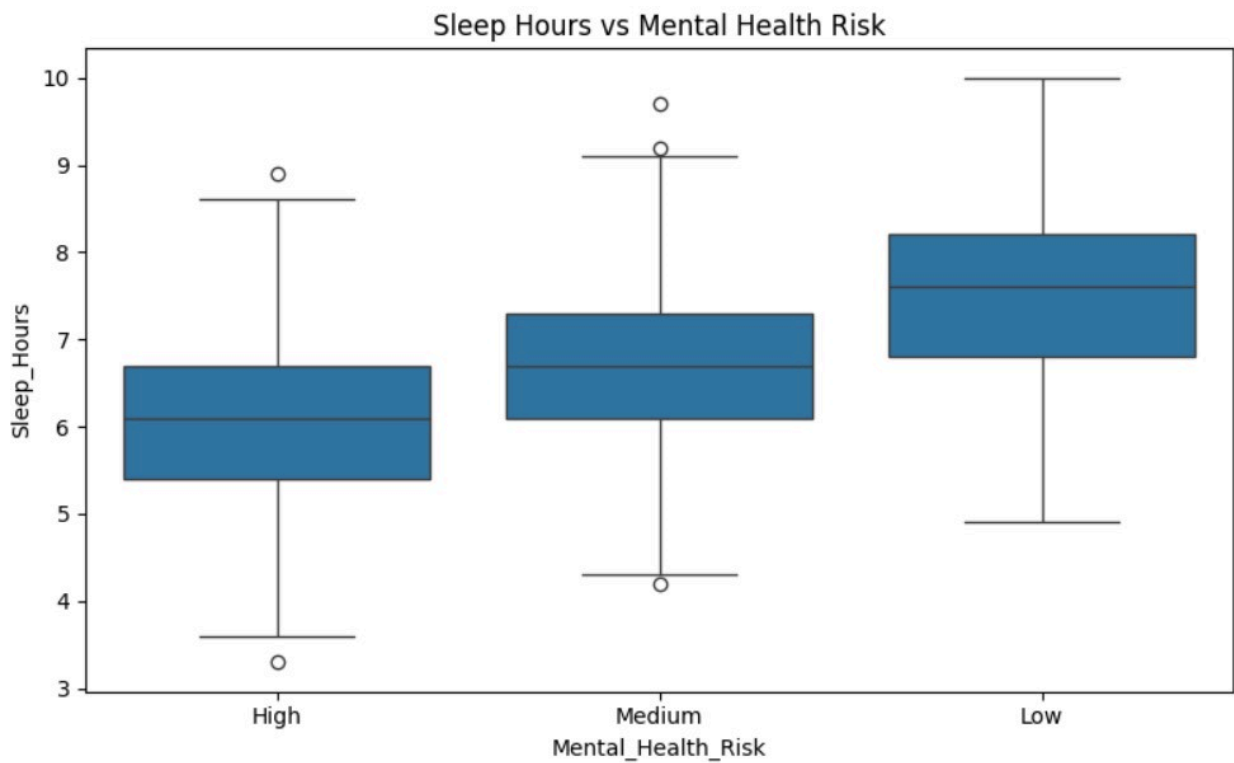


Figure 2. Distribution of sleep hours across mental health risk categories.

This visualization allows comparison of the distribution and variation of sleep hours among Low, Medium, and High-risk groups.

Financial Stress Analysis

Financial stress is another important feature examined through visualization. The notebook includes a Financial Stress Analysis graph.

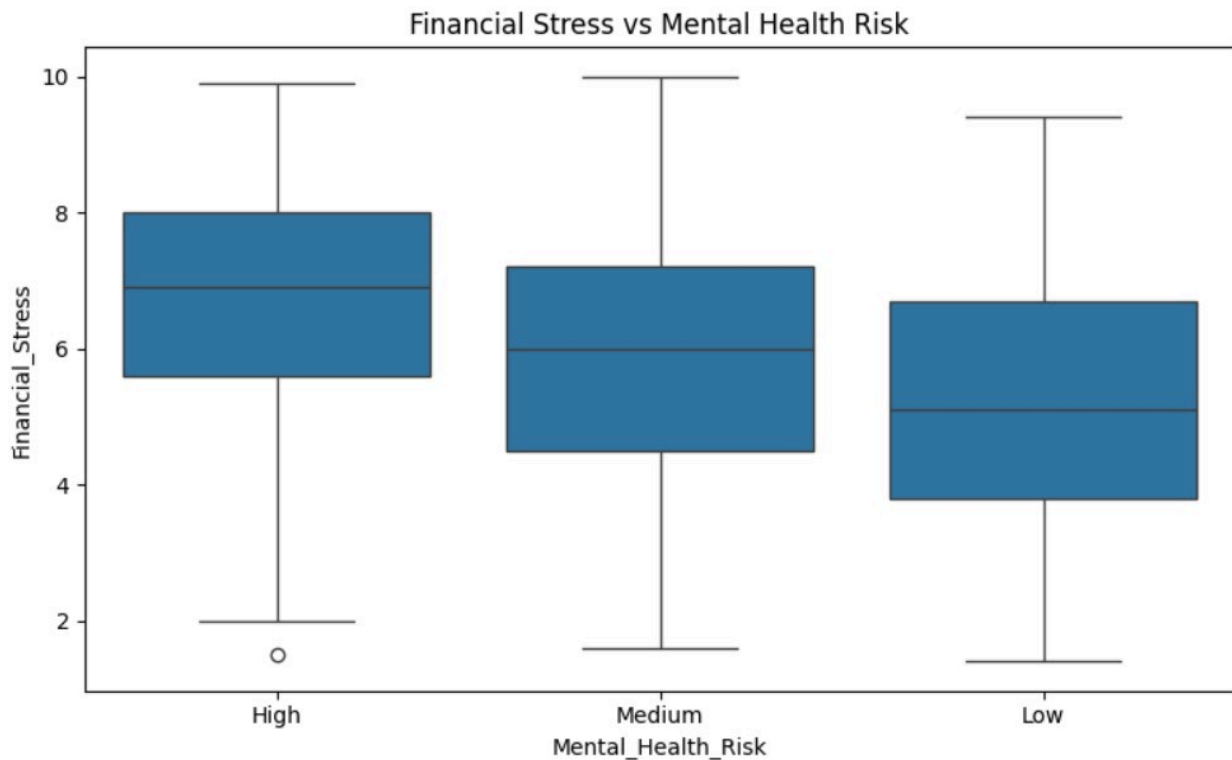


Figure 3. Distribution of financial stress across mental health risk categories.

The boxplot allows comparison of financial stress levels among the different mental health risk categories.

Correlation Analysis

A correlation heatmap is generated for numerical features. The notebook specifically creates a Feature Correlation Heatmap.

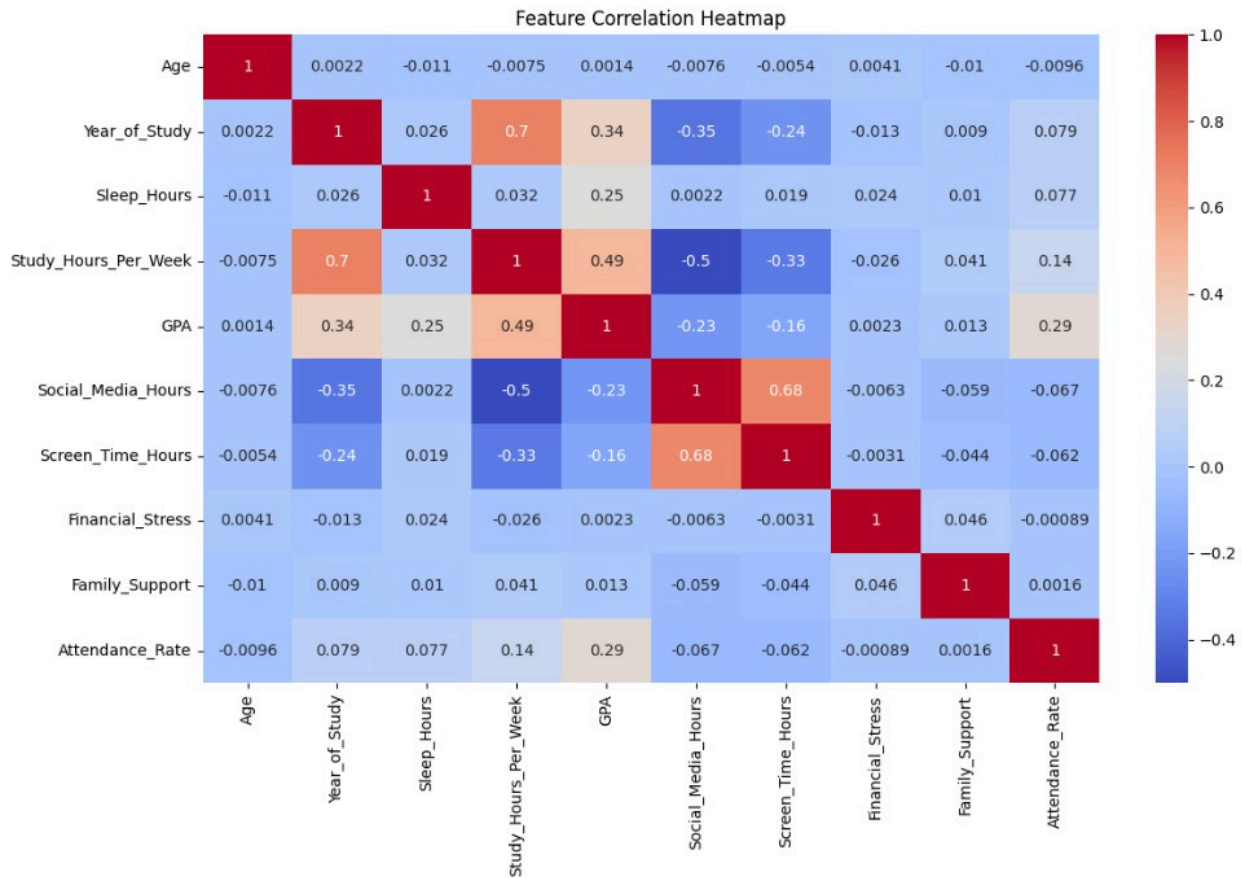


Figure 4. Correlation matrix of numerical features.

The heatmap helps identify relationships among numerical variables such as age, sleep hours, study hours, GPA, social media hours, screen time, financial stress, family support, and attendance rate.

Correlation shows the strength of association between variables, but it does not establish causation.

5. METHODOLOGY

Machine Learning Problem

The project is a supervised multi-class classification problem.

Input

The model uses student-related characteristics such as:

- Age
- Gender
- Year of Study
- Sleep Hours
- Study Hours
- GPA
- Social Media Hours
- Screen Time
- Physical Activity
- Financial Stress
- Family Support
- Attendance Rate
- Part-Time Job
- Diet Quality
- Previous Mental Health History

Output

The model predicts:

Mental Health Risk = Low / Medium / High

Feature Engineering and Preprocessing

The project separates features into numerical and categorical groups.

The numerical feature list includes variables such as Age, Year of Study, Sleep Hours, Study Hours per Week, GPA, and other numerical attributes.

The preprocessing pipeline handles the two feature types separately:

Numerical:

Missing values → Median Imputation → StandardScaler

Categorical:

Missing values → Most-Frequent Imputation → One-Hot Encoding

This approach allows the different types of data to be processed appropriately before being provided to the classifiers.

Train-Test Split

The dataset is divided into:

- 80% Training Data
- 20% Testing Data

The code uses:

- `test_size=0.2`
- `random_state=42`
- `stratify=y`

Stratification helps preserve the target-class distribution between the training and testing datasets.

6. MACHINE LEARNING MODELS

Five classification algorithms are implemented.

Logistic Regression

Logistic Regression is implemented with:

`max_iter = 2000`

The model is combined with the preprocessing pipeline.

K-Nearest Neighbors

KNN is implemented using: `n_neighbors = 5`

Decision Tree

The Decision Tree classifier is implemented with:

```
random_state = 42
```

Random Forest

The Random Forest classifier uses:

- `n_estimators = 200`
- `random_state = 42`

Support Vector Machine

The SVM classifier uses:

- `probability=True`
- `random_state=42`

IMPLEMENTATION

Programming Language

The project is implemented using:

Python

The notebook uses a Python 3 environment.

Libraries

The project uses:

- Pandas – data loading and manipulation
- NumPy – numerical operations where required
- Matplotlib – plotting

- Seaborn – statistical visualization
- Scikit-learn – preprocessing, model training, and evaluation
- OS – output directory creation

The notebook imports Pandas, Matplotlib, Seaborn, and OS for the data analysis and visualization stages.

Model Training

Each model is trained using the training dataset:

`X_train, y_train`

After training, predictions are generated on:

`X_test`

The notebook then calculates the accuracy using `accuracy_score`.

8. RESULTS

Model Performance

The actual results obtained from the notebook are:

Model	Accuracy
Logistic Regression	75.50%
SVM	71.75%
Random Forest	69.25%
KNN	57.00%
Decision Tree	56.75%

These are the actual test accuracies produced by the project code.

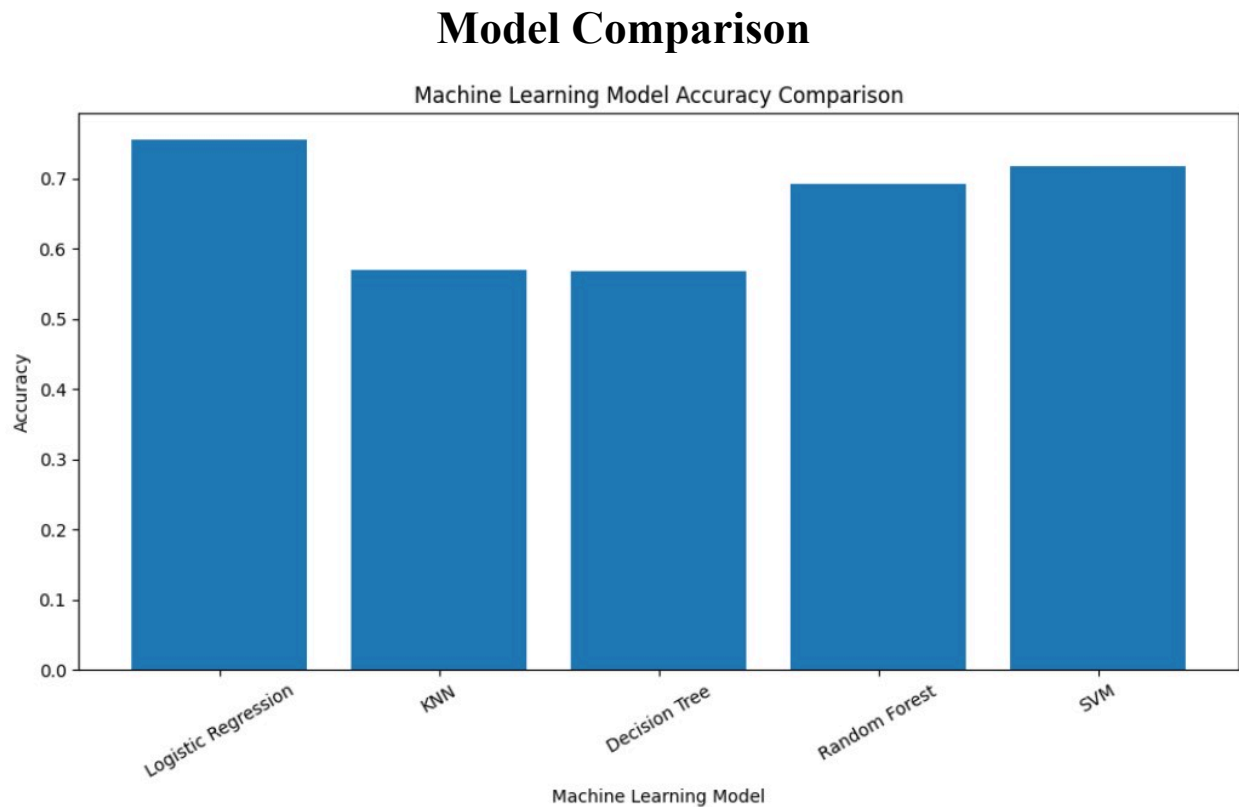


Figure 5. Comparison of test accuracy achieved by the five machine learning models.

The graph shows that Logistic Regression achieved the highest accuracy, while Decision Tree achieved the lowest accuracy.

The ranking is:

1. Logistic Regression — 75.50%
2. SVM — 71.75%
3. Random Forest — 69.25%
4. KNN — 57.00%
5. Decision Tree — 56.75%

The model comparison graph is generated directly by the notebook.

DISCUSSION

Interpretation of Results

The results demonstrate that the five machine learning algorithms have different levels of performance on the student mental health dataset.

Logistic Regression achieved the highest test accuracy at 75.50%. This indicates that it correctly classified a large proportion of the test samples.

SVM was the second-best model with an accuracy of 71.75%, followed by Random Forest at 69.25%.

KNN and Decision Tree performed considerably lower, achieving 57.00% and 56.75%, respectively.

Therefore, based on the evaluation metric implemented in this project, Logistic Regression is the best-performing model.

EDA Findings

The dataset contains a mixture of academic, lifestyle, demographic, and social features.

Missing values were found in Sleep Hours, GPA, Social Media Hours, Financial Stress, Family Support, Attendance Rate, and Diet Quality.

The target classes are reasonably balanced, with 680 High-risk observations and 660 observations in both Medium and Low categories.

The project also explores sleep hours, financial stress, and correlations between numerical variables through visual analysis.

Challenges

The major challenges addressed in the project include:

1. Missing values in the dataset.

2. Mixed numerical and categorical data.
3. Need for categorical encoding.
4. Different numerical feature scales.
5. Comparing multiple classification algorithms.
6. Maintaining the target-class distribution during train-test splitting.

The preprocessing pipeline addresses these challenges systematically.

Limitations

The current project has some important limitations.

Limited Evaluation Metrics

The notebook evaluates the models using accuracy only. It does not calculate precision, recall, F1-score, or ROC-AUC.

No Confusion Matrix

A confusion matrix is not included in the current implementation. Therefore, class-specific prediction errors cannot be examined.

No Cross-Validation

The project uses one 80/20 train-test split rather than k-fold cross-validation.

No Hyperparameter Tuning

The project uses selected model parameters rather than GridSearchCV or RandomizedSearchCV.

No Feature Importance Analysis

The current implementation does not calculate feature importance or determine which variables contribute most strongly to the predictions.

Sensitive Nature of Mental Health Data

Mental health risk prediction is sensitive. The model should not be treated as a medical diagnosis or a replacement for professional assessment.

10. CONCLUSION

This project developed a complete machine learning pipeline for predicting student mental health risk using demographic, academic, lifestyle, social, and behavioral information.

The dataset contains 2,000 student records and 17 columns. The analysis included data exploration, descriptive statistics, missing-value analysis, target distribution analysis, sleep-hours analysis, financial-stress analysis, and correlation analysis.

The preprocessing stage removed the student identifier, handled missing values using appropriate imputation strategies, standardized numerical variables, and converted categorical variables using one-hot encoding. The dataset was then divided into training and testing sets using an 80/20 stratified split.

Five machine learning algorithms were evaluated:

- Logistic Regression
- K-Nearest Neighbors
- Decision Tree
- Random Forest
- Support Vector Machine

Among these models, Logistic Regression achieved the highest accuracy of 75.50%. SVM achieved 71.75%, Random Forest achieved 69.25%, KNN achieved 57.00%, and Decision Tree achieved 56.75%.

Therefore, Logistic Regression is the best-performing model in this project based on test accuracy.

The machine learning analysis successfully demonstrates the use of multiple classification algorithms for student mental health risk prediction. Among the tested models, Logistic Regression provides the highest test accuracy of 75.50% and is therefore the best-performing model in the current experiment.

RECOMMENDATIONS AND FUTURE WORK

The following improvements can be made in future versions of the project:

1. Apply k-fold cross-validation to obtain more reliable performance estimates.
2. Use GridSearchCV or RandomizedSearchCV for hyperparameter tuning.
3. Add a confusion matrix to analyze errors for Low, Medium, and High risk classes.
4. Calculate precision, recall, and F1-score.
5. Add ROC-AUC analysis for multi-class classification.
6. Perform feature importance analysis.
7. Test additional algorithms such as Gradient Boosting or other ensemble methods.
8. Perform feature selection to identify the most useful variables.
9. Use learning curves to investigate overfitting and underfitting.
10. Test the final model on an independent dataset.
11. Apply appropriate privacy and ethical safeguards when dealing with mental-health-related data.
12. Treat the model as a screening/analytical tool rather than a medical diagnostic system.